

Update Day, 22nd June

Applied Cognitive Psychology

June 21, 2022

Abstract book

In this book you will find all abstracts of (poster) presentations of the Applied Cognitive Psychology Update Day of June 22nd. To the right of each abstract is indicated whether the project was an internship, a thesis or combined internship/thesis project. Information on supervisors (including external supervisors when applicable is also provided).

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Abstracts

What are the effects of positive and negative reviews on the online purchasing decisions for younger and older adults for various product categories?

Thesis

Joost Lanjouw

Supervisor: Krista Overvliet, External supervisor: Anouk Butterlin, Location:
Accessibility, Utrecht

Online shopping is more relevant than ever before. Online retail is an ever growing market and the corona virus has only sped up its growth. Similarly, more and more older adults (65+) are using the internet to make online purchases. In this research, I investigate what effect average ratings and user reviews have on purchasing decisions for various product categories.

Heightening conspicuity in visually impaired pedestrians to increase safety at night

Thesis

Ype de Boer

Supervisor: Albert Postma, Location: Utrecht University

At night it is harder to perceive others and when others are unable to perceive you their conspicuity becomes even more important. This master thesis researches ways in which visually impaired pedestrians can best increase their conspicuity by testing different possibilities and interviewing the visually impaired. The test existed of having an actor try to cross the street in different conditions to measure the behaviour of the cars. The results show that it does help to use a medium to increase conspicuity versus no medium, but which medium does not significantly matter. The interviews show that the visually impaired prefer low effort solutions that are subtle yet effective. They do not want to stand out too much, but they do feel the need to be noticed. The option they deemed best was adding lights to their cane, preferably built in.

Facing Our Standards: On the validity of standardised faces for perception research

Thesis

Friso Siemensma

Supervisor: Ignace Hooge, Location: Utrecht University

There has been a trend in the last decades in perception research of using standardised facial stimuli. Advances in computer graphics have been allowing manipulation over stimuli to such a degree that it has become very appealing to standardise facial stimuli in ways that remove possibly confounding factors or distractors from images, as this makes effects in perception research easier to isolate. While standardisation allows for greater control in experimental design, its inherent artificiality may limit the scope and possibilities of research which technological advances in eye tracking allow for nowadays. This thesis aims to investigate the aforementioned trend and to identify the reasoning for employing standardised stimuli through a literature study. In addition, a data analysis will be conducted using data from the 2018 Face It eye tracking experiment, which employed almost exclusively non-standardised facial stimuli, to see what results non-standardised face stimuli yield compared to standardised ones.

Eye-tracking Without Screens: Potential Problems and How To Deal With Them

Thesis

Stein Bellers

Supervisor: Niilo Valtakari, Location: Utrecht University

Any eye tracking set up has pros and cons and knowing the advantages and disadvantages of using a certain set up and how best to calibrate it is important for researchers choosing a setup for their study. A lot of relevant questions remain unanswered like how many calibration points should be used and how they should be placed. Also, whether to use binocular or monocular data and if the parallax error is constant and predictable enough to correct for it. To answer those questions, a screenless headboxed setup with custom built offline calibration software was used to record the gaze of both eyes of participants while they looked at several calibration and validation points. This gaze data was used to determine the way to calibrate this setup that results in as accurate and precise data as possible and investigate the effect of binocular and monocular data on the accuracy of the results to determine which is better.

Testing the Effectiveness of Personal Names for Alerting Transitions in Semi-Automated Cars

Thesis

Jiaxin Xu

Supervisor: Chris Janssen, Location: Utrecht University

This thesis investigates how a speech alert that uses a driver's personal name affects the speed and accuracy of responding to transitions of control in the context of semi-automated driving. Preceding studies have shown that a personal name can be an effective stimulus to capture attention. However, studies using personal names in speech alerts in cars are limited and showed mixed results. I conducted an experiment with a simulated semi-automated car in which participants were watching driving videos while playing a mobile game. Occasionally the participants heard alerts that started with either their personal name or a warning sound. They were required to immediately disengage from the game and respond to an alert by pressing a button on the steering wheel, or by pressing the brake. Results showed that participants' response times were significantly faster when hearing an alert containing their own names. There was no difference in subjective experience between conditions found. The results suggest that the use of one's name is a useful addition to an in-car alert and warrants deeper investigation for future research.

effectiveness of street crossing techniques for the blind and visually impaired in the dark

Thesis

Tyren Romijn

Supervisor: Albert Postma, Location: Utrecht University

There are about 250.000 people with a visual impairment in both eyes in the Netherlands according to the organization het Oogfonds. These individuals are considered blind or visually impaired. Humans are visual creatures and not being able to see can make everyday tasks a lot more difficult. One task which becomes more difficult is crossing the street. Not being able to see when you have to wait for a car could lead to some dangers situations. This becomes even more dangers when it is dark because drivers will see the blind individual a lot less. For this reason, street crossing techniques were developed to help them be more visible. This experiment will look at 2 techniques: biomotion and a light up cane. Participants will be put in a driving simulator and will see one of the two techniques and we will record if they stopped for the blind pedestrians.

Human behaviour in Cooperative Intelligent Transport Systems (C-ITS)

Internship

Sara Pitkänen

Supervisor: Chris Janssen, External supervisor: Rick Schotman, Location: Goudappel, Amsterdam/Remote

Cooperative Intelligent Transport Systems (C-ITS) can be useful, for example in making driving safer and more economical. The way people react to these systems could influence the overall traffic. Previous work suggests that the different behavioural aspects are not fully understood, and the attempts to simulate the behaviour are lacking. This review focused on the literature found related to behaviour and C-ITS traffic light assistants; time to green/red, green light optimal speed advisory, and countdown timers. The aim was to gather information on behavioural changes, interactions between different users, and see what benefits or disbenefits these systems might have for different road users. This information could be useful when simulating the traffic and trying to make predictions and decisions about the use of these systems. The results are controversial as there are both positive and negative outcomes when it comes to behaviour and interactions between different road users. Overall, based on the findings these assistants may have positive effects such as decreased travelling time. However, the negative effects, such as the increased risk of accidents may overcome the positive findings.

Internal Service Chatbots: Understanding employees' intention to use a chatbot in the workplace.

Thesis

Anna Praat

Supervisor: Daan van Rooij, External supervisor: Cindy van Dalen, Location: Ministry of Defence, Utrecht

Employer unattractiveness is shown as one of the main factors for employee deficiencies in the already very tight labour market in the Netherlands. One way for an employee to become more attractive is through high internal service quality. With the digitalisation of the workplace using a chatbot shows great potential to increase the quality of HR, IT and facility services. Literature points out the importance of taking into account human behaviour and emotions when implementing a chatbot in the workplace. Through the use of the Technology Acceptance Model and the Theory of Planned Behaviour, this thesis studies the intention of employees to use a chatbot for internal services. Additionally, the effect type of question asked has on the relation between attitude and intention studied.

The use of verbalization and its effect on the performance of Visual Working Memory

Thesis

Yasaman Javadi

Supervisor: Andre Sahakian, Location: Utrecht University

Imagine if our brain has multiple workspaces, and these workspaces are where the information first enters our brain. Visual Working Memory (VWM) is the workspace in our brain where the visual information becomes stored. Even though VWM's capacity is limited, and we do not consume most of it all the time, there might be useful methods to use to help our VWM perform better. What this study suggests is that maybe with the help of verbalization, the act of naming an item in a descriptive way, we can improve the performance of VWM. Our findings report that when needed visual information is not easily available in the external world, people use different methods to store those needed visual information in their VWM to do their tasks faster. And If information is not necessary to know and memorized, they are more willing to do the required task done and not consume much of their memory.

Let's break the ice: an experimental investigation of the effect of team building on human-robot interaction

Thesis

Yen-Yun (Catherine) Liu

Supervisor: Ruud Hortensius; Peter Ruijten-Dodoiu (external supervisor, Eindhoven University of Technology), Location: Utrecht University

We experimentally evaluated the influence of a team building intervention on people's behaviors and perceptions of a socially interactive robot in the short and long run. As the field of mobile robotics advances, autonomous robots are able to perform various tasks in the setting of social interactions. However, while people have great expectations of the benefits that robots can offer to society, much work remains to be done to increase robots' social effectiveness in real-world situations. As a result, we investigated the impact of team building based on a human-to-human interaction script. Sixty-one participants took part in a problem-solving quiz game, with half receiving the intervention and the other not. Both groups completed two sessions with four to eleven days of zero exposure to the robot in between. A significant main effect of team perception was found, with people who reported a higher score of team identity, perceived the robot to be warmer and more competent and the interactions to be more engaging. They were also more willing to use the robot in the future. The time and the intervention showed no effect. This study found that team perception is critical in predicting people's behaviors and perceptions of a socially interactive robot, which can inform developers in this field that forming a team identification could be beneficial to start off a positive human-robot relationship.

Expectations vs. reality: What is the influence of knowledge framing on the perception and understanding of a robot?

Thesis

Susanna Marie Spörl

Supervisor: Dr. Ruud Hortensius , Location: Utrecht University

Expectations about robots, which are mostly shaped by media and science fiction narratives, are often vague, undefined, and unrealistic. Based on rare interaction possibilities with robots in everyday settings, the question arises about what influence expectations have on the perception and understanding of robots. According to Dennett (1971), we explain and predict our surroundings with different strategies. Recent research shows that people make references to mental states and can adopt the intentional stance to explain the behavior of a humanoid robot. It was suggested that prior expectations influence which strategy people adopt. This gains relevance when considering the consumer market, where advertising often contains mentalistic statements. The present study examines how framing a robot affects preferences for mechanistic or mentalistic statements, anthropomorphism, effectance, and mind perception. Preliminary results indicate that no general effect of framing on the variables studied could be found. Possible contributions and influencing factors are further discussed.

Top-down Influence on Skin Conductance Response

Thesis

Sander van Wincoop

Supervisor: Christoph Strauch, Location: Utrecht University

An effect has been found that pupil size is influenced by a higher level visual analysis of what is perceived, rather than just reflexive and based on lower level features. In my thesis I am conducting an experiment to investigate if a similar higher level visual analysis also plays a role for skin conductance. This is done by comparing skin conductance of participants when showing them images correlated to high temperatures and comparing this to their skin conductance while showing them images correlated to low temperatures. If this higher level visual analysis is indeed also present in skin conductance we would expect to see a significant difference between categories.

What is the role of response inhibition in mediating the associations between lifestyle, impulsivity, and mental health? How does gender affect these associations?

Thesis

Margherita Ferrero

Supervisor: Daan van Rooij, Location: Utrecht University

The role of diet and other lifestyle factors such as physical exercise on mental health has been well established. Especially the influence of such factors on the alleviation of depression and other internalizing disorders has been well established (Lopresti, 2013). However, the relation between diet/exercise, mental health disorders (like ADHD) and gender has been a lot less researched. In recent years, several studies have shown an association between diet and lifestyle factors and impulsivity in the general population (Schworen et al., 2021) and a positive association between healthier diet, cognition and gender (Guo et al., 2021). It is not yet clear through which (neuro)biological channels this association between diet/lifestyle and impulsivity is mediated. A recent study by van Rooij et al. (2021) has shown a small mediation of the effect of diet on impulsivity through the thickness of frontal-parietal brain areas, where response inhibition also underlies. Therefore, we might be able to explain the effects of diet on impulsivity further by investigating the role of cognitive control within this causal pathway and how men and women might present different results.

De invloed van automatisering van voertuigen op controlekamers. Een benchmark van de spoor-, metro- en luchtvaartsector.

Internship

Floris Geerts

Supervisor: Sjoerd Stuit, External supervisor: Alexander Admiraal, Location: ProRail, Utrecht

Om capaciteitsverhoging op het spoor te ondersteunen, gaat er geautomatiseerd worden. We hebben een literatuuronderzoek gedaan naar de invloed van automatisering van voertuigen op controlekamers. Een lage werkdruk kan zorgen voor een lagere situation awareness. Om dit te onderzoeken hebben we niet alleen gekeken naar literatuur binnen de treinsector, maar ook naar andere sectoren: de metro en luchtvaart. Vondsten vanuit deze sectoren hebben we vertaald naar de treinsector. We hebben gevonden dat de capaciteitsverhoging de werkdruk voor de treindienstleider verhoogt. De gepaarde automatisering om deze verhoging te ondersteunen verlaagd juist de werkdruk, omdat treinen voorspelbaarder rijden en door een vermindering in menselijke fouten. Zowel in de luchtvaart als bij de metro is het systeem in de controlekamer meer geïntegreerd. Dit zou ook aan te raden kunnen zijn voor de treinsector, wanneer er meer wordt geautomatiseerd.

I wrote a systematic review about the history of pupil perimetry, a novel method to detect visual field defects.

Thesis

Tom van Kempen

Supervisor: Marnix Naber, Location: Utrecht University

For decades, assessment of visual function has been conducted using a subjective perimetry test which exists of a spherical apparatus presenting bright stimuli at various regions of the visual field (VF). The subject subsequently indicates whether the stimulus is observed, which results in an independent measure of sensitivity of each retinal region. This standard automated perimetry (SAP) lacks objectivity due to the dependency on the subjects' responses. Pupil perimetry is a novel method that utilizes the pupillary light reflex (PLR) to objectively map the sensitivity of the retinal regions. By presenting a stimulus at a specific location, the extent of contraction of the pupil functions as a measure for sensitivity of that region. The independency of the subjects' responses creates opportunities for young children or other patients that lack task-comprehension, motivation, attention, concentration or verbal skills, essential factors for accurate SAP measurements. In the last 60 years, research on pupil perimetry has expanded. Various methods, apparatus and patient groups have been subjected to the objective VF assessment method and given the advantages towards the currently used SAP, implementation into ophthalmological clinics seems close. Despite the wide range of applications of pupil perimetry, a review has yet to be published. To support future research into pupil perimetry, this chronological narrative review presents the most important milestones, methods and discoveries in the history of the matter. By accurately screening all publications surrounding the concept of pupil perimetry, this review will summarize the developments in the last 70 years and visualize the different tested clinical groups, apparatus and stimulation paradigms.

How communication can influence task performance on a collaborative puzzle task

Thesis

Ellen Verbunt

Supervisor: Roy Hessels, Location: Utrecht University

Human-robot interaction (HRI) has made a huge increase in the last decade. Because of the increasing spaces that are being shared with robots, intuitive and safe interfaces are essential. The least developed type of interaction with robots is collaboration. To create good HRI, it should be very similar to human-human interaction. As a start to the goal of achieving a good collaborative interaction between humans and robots, it is important to observe human-human collaboration. For my thesis I created a collaborative puzzle task, in which a pair of participants were asked to copy figures made of Duplo while wearing eye-tracking glasses. Half of the time they were allowed to talk with each other, and the other half of the time they were not. From this data, I'll hopefully be able to predict how verbal and nonverbal communication influences the task performance, and what the gaze patterns are.

Social robotics model for children

Internship

Sebas van Hal

Supervisor: Roy Hessels, External supervisor: Heqiu Song, Location: TU, Eindhoven

For my internship I was asked to help my supervisor Maggie, with creating a social robotics model for children. She is currently working on her PhD, which involves using a social robot as a music teacher for elementary school children. During her studies she discovered that no social robotics model and questionnaire for children existed yet, and that many of the models that are currently used for technology acceptance and/or social robotics are severely lacking in the psychological aspect. Thus, the aim of the internship is to work together in creating an actual social robotics acceptance model for children, which she (and future social robotics researchers) can use for her studies. We also aim to validate the model we are creating immediately, which is why my internship will continue during the summer. We are currently still working on creating the model, and what aspects to include or exclude and how to make the questions suitable for elementary school children.

Working memory performance under multitasking conditions

Thesis

Bjorn van Vliet

Supervisor: Chris Klink, Location: Utrecht University

Literature proves that dual task interference (DTI) decreases proficiency on simple tasks. One of the ways that we perform dual tasks is while navigating. While navigating it is of importance that the driver remembers the traffic signs that are passed in order to correctly find the wanted route. This is a roll that is fulfilled by the working memory. Research stated that the limit of our working memory is somewhere 9 to 3 chunks. Therefore this research will try to find out how working memory performance is influenced by performing a navigating task simultaneously. The research question that will be answered is: “what is the influence of a navigation task on working memory capacity”. To perform this study a simple driving task will be created in combination with a memory task. Significant effects were found for number of signs, the driving task, number of collisions, age, experience and the interaction between number of signs and the driving task and the performance on the memory task.

Algorithms to Denoise eye tracking data - an attempt to improve classifier performance

Floris van den Esschert

Supervisor: Roy Hessels, Ignace Hooge, Location: Utrecht University

Combined
intern-
ship/thesis

Eye tracking represents a technique which provides information about eye position and movement over time. When humans visually scan scenes they often do so using a pattern of fixations and saccades. The increased complexity of eye-tracking signals (due to equipment or human error) can increase the amount of noise inherent in the data (i.e. variability in gaze position during fixations or periods of data loss). Ideally the magnitude of noise in the eye-trackers output should be low enough so that it does not obscure relevant measurable events. Simple classifiers can detect these events well, however these classifiers seem to do worse the higher the amount of noise in the data. In this paper we focus on implementing different algorithms (Savitzky-Golay, Total-Variation Denoising and Kalman Filters) that can denoise the data and test their effectiveness by comparing the denoised data to the raw data in classifier performance.

Does transparency pay off? Effect of wait duration and filler interface on wait experience in the context of P2P payments

Thesis

Monica Rapisarda Botero

Supervisor: Dr. Irene van de Vijver, External supervisor: Simon Thorsander, Location: Verisure Innovation AB

Various strategies have been formulated to manage online waits, such as displaying temporal indicators (e.g. progress bars), distractor cues (e.g. animations), or a running tally of the system's operations (i.e. operational transparency). In this online study, we investigated whether making users wait engenders more negative evaluations compared to delivering instant results, as well as the effect on user experience of operational transparency in comparison with other common wait filler interfaces. We prototyped a peer-to-peer (P2P) payments web application and assigned 567 participants to either one of nine waiting x wait filler interface conditions or to an instantaneous (no wait/ no filler) condition. We found that users are overall tolerant of short to moderate waits and do not appear to expect or reward instantaneous service delivery; however, longer wait durations yield increasing uncertainty and decreasing enjoyment. Additionally, we found operational transparency to be less impactful than expected on the user experience. We propose that this finding may be the result of multiple factors, including the changed technological context since the introduction of the concept in literature as well as consumers' expectations for P2P payments services.

Optimizing Inspire ERP Software Data-Reporting with International Business Communication Standards (IBCS).

Internship

Barry Kleijer

Supervisor: Bauke van der Velde, External supervisor: Douwe van Dijk, Location: Den of Data

Inspire is an ERP Software supplier for rental companies. Den of Data is a business intelligence firm that was consulted by Inspire to optimize their current ERP dashboard. Standardization is common in areas like traffic signs, music and road maps but not for presentation of business data. This leads to confusing communication and poor decision-making. The current dashboard is low on information density and uses less optimal visuals like pie charts and excessive color use. After identifying the needs of all stakeholders, the current dashboard was rebuild with restructured data from a SQL database and imported into Microsoft Power BI for further visual optimization with IBCS. After applying IBCS, results can be interpreted throughout all layers of the organisation without additional processing. The dashboard is responsive now and shows results in a clear and consistent way, enabling faster and better understanding of business performance.

The factors influencing Aesthetic preference in terms of ancient Chinese paintings

Thesis

Ziyu Liu

Supervisor: Stuit, S.M. (Sjoerd), Location: Utrecht University

Abstract Objectives: The aim of this thesis is to find out how and in what way would low-or high-level image features, as well as cultural effects influence our aesthetic preferences. This study is interested in comparing the aesthetic preferences of people from the eastern and western countries in ancient Chinese paintings. **Method:** This study will analyse the like and dislike images in relation to their low-and high-level image features and cultural factors. Participants are with normal or corrected-to-normal visual acuity and colour vision. The data were collected from both in-lab (N = 50) and online participants (N = 500) using an online questionnaire with images of ancient Chinese paintings (Year 550 -1900) with a short exposure (1s) and long exposure image presentation time (3s) presented on a computer screen. Each of the in-lab participants judged all paintings (N=800) and online participants (N=100) on a binary scale without repetition. This study used machine learning as a tool to find image features that have predictive value for the category to which the image belongs (N=800). **Results:** Not yet ready. But literature suggests cultural effects in terms of personal identity and low-or high-level features would influence our aesthetic preferences.

The subitization of multiple items across multiple fingers.

Thesis

Zeinab Lalaoui

Supervisor: Krista Overvliet, Location: Utrecht University

In 1871 it was discovered that the human brain could enumerate up to 3 items at once. This type of enumeration is called subitizing. Previously it was thought that subitizing only occurred in vision, but studies in tactile enumeration have shown that subitizing also occurs through touch. This study will investigate whether subitizing still occurs across fingers when each finger is presented with multiple items and whether this increases the subitizing range. To answer the question 1 experiment with two conditions was conducted. In the first condition of the experiment the participants were presented with the same number of dots across fingers. The same fingers were used on both hands. During the second condition the same fingers were used on both hands as well. However, each finger was presented with a different number of dots.

National Brands (A-brands) vs. Supermarket Brands (B-brands): Detecting bias in binary preference-based food decisions in the Netherlands”

Eleni Evangelatou

Supervisor: Dr. Martijn Mulder, Location: Utrecht University

Combined
intern-
ship/thesis

The influence of branding on consumers’ choice behavior has been substantial. However, the psychological mechanisms underlying this influence have yet to be discovered. Initially inspired by the previous work of Piliastides, M. G., & Ratcliff, R. (2013), we aim to investigate whether a bias is introduced in consumers’ choice behavior when making a binary choice between labeled food items in a typical Dutch supermarket, and measure it. Under time pressure, a two-alternative forced choice test (2AFC) is conducted that pairs two main brand types of food products, namely National Brands (or A-brands) and Supermarket Brands (or B-brands). A-brands are famous and established products manufactured and developed by a company and are distributed nationwide or internationally (e.g., Coca-Cola), while B-brands are owned and labeled by retailers and are exclusively available in their stores (e.g., Albert Heijn Cola). During the next period of my thesis, we will use computational modeling (Drift Diffusion Model) to analyze our data and either confirm or reject our hypothesis that branding itself affects choice behavior.

A Machine Learning Way to predict the effects of image features in the emotional faces recognition

Thesis

Wenkai Lang

Supervisor: Sjoerd Stuit, Location: Utrecht University

Background: Human conveys information through facial expressions in the process of interpersonal communication. In some early investigations of expression detection, both anger superiority effect and happiness superiority effect are reported. In order to resolve the contradictory reports of superiority effects in expression detection, some studies had put forward that these findings implied that expression detection is feature-based Methods: In this project, a machine learning method is used to verify if the reported inconsistencies found in the previous line of research are predictable by the low-level image feature. An online visual search task was conducted with stimuli from NimStim face database. These face images have been manipulated in four different ways: matched original one, masked images with noise, images with only significant features and images without all significant features. 30 participants were required to determine whether all the faces presented are the same emotions or if there is any different expressions among a series of presented nine-image arrays. by pressing the corresponding keys. Results: All data are processed and analyzed by Matlab. After performing an analysis of comparing the expression-related feature differences among four different conditions, the predictive value of low-level image features for expression recognition is confirmed. It suggested that .

Human Visual Perception in Role of Driverless System

Thesis

Nandi Sihol

Supervisor: Surya Gayet, Location: Utrecht University

The size of an object and its appearance in a logical setting have led to incorrect human perceptions. This research consists of two experiments, the first of which seeks to identify a blurred object, and the second of which requires participants to accurately report the size of an object on the stimulus. Consequently, the focus of this study has been on detecting objects in congruent and incongruent circumstances and estimating the sizes of objects that appear in faraway and near proximity. Ultimately, the findings of this research will be used to improve the autonomous driving system in modern automobiles, which has trouble recognizing its surroundings.

Trust in gamified persuasive technology

Thesis

Laura Cibulskyte

Supervisor: Ana M. D. Ciorcalan, Location: Utrecht University

Technology is starting to play a key role in our everyday life. The application of emerging technologies vary from entertainment, and education to clinical diagnosis and interventions (Aldenaini et. al., 2020). The specific type of technology that is discussed in the current project is persuasive technology. Persuasive technology has a broad application. There can be found examples in the physical or mental health applications, software encouraging people to act more environmentally friendly, educational applications, and many others (Aldenaini et. al., 2020). Persuasive technology shows to increase users' motivation and can help in forming healthier habits. But in order to adapt the technology and use it at its full, one of the deciding factors is trust. In this project, the question of what it takes for a persuasive technology to be trusted is investigated. After executing qualitative research it was found that Trust is a multidimensional concept that can be explained by the times such as Technology is transparent and functional, external factors, emotions, internal factors, and design. Each of the themes contains of subcategories that can explain the mental models of deciding factors.

Fear of Climate Change and its effect on Food Choice in young adults

Thesis

Gianfranco Bocanegra

Supervisor: Monique Smeets , External supervisor: Patricia Bulsing, Location: De Haagse Hogeschool

Even though there are plenty of studies examining the impact of fear appeals from the health and marketing disciplines, fear appeal and their effect in changing diets, evoking environmental reasons on young adults are limited.

The purpose of the study is to determine the usefulness of the emotion fear in order to produce a change in intention and behavior to a “plant-based” diet among young adults in the United Kingdom, evoking environmental reasons.

Reading Emotions: Automatic Recognition of Engagement Using Facial Expressions and Physiological Measures

Thesis

Fatih Celalettin Deniz

Supervisor: Stella Donker, External supervisor: Tess den Uyl, Location: noldus information technology, Wageningen

Engagement is a state of being involved, occupied, or fully absorbed in something, the advantage of engagement is widely known in the literature. Despite its importance, there are many methods for measuring engagement. This study utilizes both subjective and objective measures to investigate engagement by using facial action units, skin conductance, eye movements, and pupil diameter to understand how these measures are affected by engagement. Furthermore, machine learning models were used to classify engagement using action units. The results indicated that there were some differences when participants are engaged, compared to neutral and not engaged, but effect sizes were low. These results suggest that no existing specific facial expression for engagement, or changes in the facial muscles are so subtle to detect.

How do Cooperation or Competition Tasks and Speech Influence Gaze Patterns in Live Interactions

Thesis

Lan Yang

Supervisor: Niilo Valtakari, Location: Utrecht University

As herd animals, humans have imperative biological needs to interact with others; social interaction involves the apparent or potential transmission of a series of social signals, such as gaze, facial expressions, speech, and gestures on a daily basis. Still, it remains unclear how human's gaze patterns change in different live social interactions. Therefore, the current study adopted a 2 (task type: cooperation, competition) $\times$ 2 (speech type: speech, non-speech) mixed design to investigate differences in gaze patterns in the live social interaction. The result demonstrated that task type had a significant effect on gaze pattern; participants presented more ordered searching paths in the cooperation task than in the competition task. However, there was no significant evidence to prove different speech conditions influence human gaze behavior.

The effect of CEPs on mental availability of brands

Internship

Anne Saray Klop

Supervisor: Anne Gerlicher, External supervisor: Marceline Veen / Noud Schartman,
Location: Validators

For my internship I got a position at Validators, a neuromarketing company. For my internship I will be analyzing pre collected data which has been collected through marketing campaigns. I will be analyzing this data for the effect of mental availability of a brand on a specific category entry point (CEP). This process will take place in three phases. The first two phases consist of tagging commercials for CEPs and the data will be further prepared for data analysis. The analysis is phase three. With the analysis the effect of mental availability and CEPs will be examined.

The influence of stress on spontaneous self-touching behaviors

Thesis

Corry Dongoxe

Supervisor: Anouk Keizer , Location: Utrecht University

Receiving touch from others is not always available and even though it is known that touch might constitute a valid strategy to deal with stress, there is remarkably not a lot of research available on the effect of spontaneous self-touching (SST) behaviors and stress. The present study sought to investigate if exposure to stress leads to an increase in spontaneous self-touch. Stress was induced by using a modified version of the Trier Social Stress Test (TSST). The task consisted of a free speech and solving an arithmetic problem respectively. The video recording was used to analyze and code self-touching behaviors before, during, and after the TSST. The results indicated that there was a significant difference in the amount of spontaneous self-touch behavior before and during the TSST and during and after the TSST. Taken together the results imply that there was more SST when participants experienced more stress.

An exploration into the effects of age on the Augmented Reality Mobility Assessment Course in healthy participants. Thesis

Lauren Travers

Supervisor: Chris Klink, External supervisor: Zalya Berger, Location: Royal Visio

Visuospatial neglect (VSN) occurs frequently after stroke and is diagnosed by neuropsychological assessment such as paper-and-pencil tasks and functional assessments. However, these tests lack sensitivity to capture all neglect subtypes. This means the tasks chosen (or not chosen) to detect VSN can be crucial in detecting (or missing) a patient's neglect subtype(s). Augmented reality (AR) offers new ways to benefit patient testing and rehabilitation after stroke. However, very little is known on the effect of age in the performance of tasks, usability, and diagnostic criterion in AR neuropsychological tasks in the 65+ group. The aim of this study is to test whether there is a significant effect of age on the performance of healthy participants in a recreation of the Mobility Assessment Course (MAC) as an AR application on the Microsoft HoloLens 2. Healthy participants will navigate through corridors, searching for 24 targets displayed through AR. Line bisection, apples cancellation tasks, and a questionnaire will also be used to compare the results of the AR-MAC between age groups. Additionally, this study will provide norm scores based on healthy participants' performances in the AR-MAC.

The Role of Media with Sexual Content on Adolescent's Sexual Risk Behavior and Pleasure During Sex Thesis

Deniz Yerebakan

Supervisor: Joyce Endendijk, Location: Utrecht University

The current research explores how various media types with sexual content influence adolescents' sexual risk behavior and pleasure and how this relationship is mediated by adolescents' sexual double standard stereotypes. Adolescents who are going through active changes in their sexuality are frequently exposed to such content. Considering this, there is earlier research indicating associations between exposure to sexual content and adolescent sexual behavior, and between exposure to sexual content and adolescent SDS stereotypes. Data were collected through an online survey of adolescents (aged 16-20 years, N=257). Regarding these associations, no significance was found. However, with no significance, the addition of adolescent SDS stereotypes in the model increased the effect size. Some individual significant correlations were identified based on the level of exposure to specific media types; (1) the level of exposure to reality TV was found positively correlated with adolescent sexual risk behavior and (2) the level of exposure to porn was found significantly correlated with adolescent sexual pleasure.

Using peer assessment in online courses: a qualitative approach to deciding which behavioural and structural factors to incorporate in MOOC design

Thesis

Janne Venderink

Supervisor: Chris Paffen, External supervisor: Yves Plees, Location: SUE Behavioural DDesign, Amsterdam

Extensive evaluation for MOOCs can be nearly impossible for teachers due to the immense amount of work. The most applied evaluation tools in MOOCs therefore comes in the form of peer assessment. This form of assessment consists of a multitude of variables, which have not always been accounted for in existing research. The present study aims to provide a clear overview of knowledge and gaps on the different variables that peer assessment is comprised of, by categorizing these over behavioural and structural factors. Specifically, a lack of knowledge exists on the behavioural factors affecting the peer assessment process and outcomes – on which knowledge is extended by empirical findings. The present study consists of two parts with two different research methods: the first part provides a structured overview of variables that peer assessment is comprised of and identifies knowledge gaps through a systematic literature review. To generate new insights on most particularly the behavioural factors, three focus groups with university students were carried out which comprises the second part of this research. This paper especially contributes to existing knowledge by unravelling the variable motivation and its potential ways to increase student motivations for thoroughly and truthfully engaging in the peer assessment process. Although this study contributes to determining how peer assessment is most optimally implemented, many controversies still exist, and optimal strategies for certain variables remain unclear. Therefore, providing a complete advice for optimal course design is not reasonable, though a clear overview with recommendations and trade-offs for each variable is given. The present study helps course designers to decide which identified variables of peer assessment to implement in the course, and remaining knowledge gaps are identified for which recommendations for future research are made.

Client engagement in a mental health institution

Thesis

Najoua Marroun

Supervisor: Stella Donker, External supervisor: Laura Bijkerk, Location: Oh My Mood, Amsterdam

In this study, we aim to investigate the experience of client engagement and its sustainment following a blended, transdiagnostic, positive psychology-based intervention for various mental health problems at a mental health institution in the Netherlands. Furthermore, we wanted to know which factors facilitated or obstructed engagement with the intervention. Engagement has several definitions in the literature; however, we will use the model of Kelders and colleagues. This model shows that engagement consists of three elements: behavioral, cognitive, and

affective engagement. To investigate how the experience was for the clients, we conducted a semi-structured interview within a homogeneous sample. We adapted a phenomenological method approach by using Interpretative Phenomenological Analysis. We interviewed four clients who were already finished with their treatment or were in the final phase of their treatment. For behavioral engagement, clients were able to be open and honest with their therapist. As for cognitive engagement, the clients were able to reach their personal goals through the intervention. Besides most of the clients could identify with the rationale of the intervention. Most clients indicated that they found the experiences of other clients helpful for them. Other facilitators were the characteristics of the therapist, the content of the intervention, and the relationship with the therapist.

Eye Tracking study for Specialty Beers at Jumbo

Internship

Sara Lutgerink

Supervisor: Roy Hessels, External supervisor: Renee Göbbels, Location: Unravel Research, Utrecht

In this study, eye-tracking was used to uncover the process of choosing specialty beers. 30 shoppers took part in this study that took place in a Jumbo supermarket. Jumbo had just sorted their specialty beers in a new way, namely by brand instead of by category. The goal was to gain insight into the importance and order of dominant choice factors. Based on the eye-tracking and survey data we advised Jumbo to consider the following: 1. Use vertical blocks for optimal brand recognition. 2. Use secondary organization by beer type. Lighter beers on the left or top, darker beers on the right or bottom. 3. Try placing luxury beers on top shelves and cheaper beers on bottom shelves. 4. Consider organizing multipacks on the bottom shelves or to the right side of the shelves. 5. Use contrasting brand labels to have brands stand out more, and have customers explore more.

Internship at Emergo by UL

Internship

Hinaho Ishikawa

Supervisor: Dr Stella Donker , External supervisor: Tamara Will , Location: Emergo by UL

This poster presents my internship experience at Emergo by UL. Emergo by UL is a consultancy firm that specializes in medical devices. I am doing my internship at Human Factors Research & Design team, which focuses on usability research and design to improve the safety and effectiveness of medical devices. As a human factors intern, I get to be involved in hands-on experiences such as writing portions of proposals and reports and learning how to conduct various analysis methods (e.g., known problem analysis).

Objectively measuring the role of spatial attention, in pseudoneglect and information gathering through pupillometry

Tijs Aarsen

Supervisor: Christoph Strauch, Location: Utrecht University

Combined
intern-
ship/thesis

Many decisions we make during a day are driven by spatial attention. For example: "Which apple do we choose in a supermarket?". Spatial attention however is not divided equally between both hemifields. The phenomenon of pseudoneglect describes an innate attentional bias towards the left hemifield. We attempt to measure this bias through a set of novel stimuli, trying to predict the role of the orienting and alerting attentional networks. These networks play a role in all forms of attention and are not limited to pseudoneglect. While either overtly or covertly attending a stimulus, we slowly gather information before coming to our eventual decision. This process of information gathering is described in the Drift Diffusion Model (DDM). The DDM uses RT and accuracy to predict the rate of information gathered in a two-alternative forced choice paradigm. Both these phenomenon can be tested through pupillometry. If we are able to predict the rate of information gathering from the pupillary response, we could be able to argue that spatial attention and the pupil interact as early as the brainstem (locus coeruleus).

Does positive far transfer of motor skills occur with extensive training? The effect of instrumental training in musicians

Nina Hoogerland

Supervisor: Lara Koijsk, External supervisor: Rebecca Schaefer, Location: Leiden University, MBHT lab

Combined
intern-
ship/thesis

The main aim of this thesis is to examine whether musical training positively affects far transfer in the motor domain. For this purpose, we compared the performances of 24 musicians with instrumental training to 14 musicians without such training and non-musicians on the pursuit rotor task. Participants were grouped based on the Gold-MSI and additional questions on practice habits. Both groups showed clear improvements on the pursuit rotor task, and their learning rates were comparable. Of significance was the fact that instrumental training positively affected the performance on the first trail of the motor learning task and, therefore, the overall performance of the musicians with instrumental training. These results suggest that positive far transfer of a previously trained motor skill to a different context occurs. Moreover, these findings support previous research indicating that musical training could help achieve positive far transfer of motor skills. For future research, it would be interesting to look at more difficult motor learning tasks, since the participants reached their performance plateau early in the trials.

Effect van een aardvast referentiekader op cybersickness bij bewegende beelden in VR

Thesis

Rianne van Miltenburg

Supervisor: Chris Paffen, External supervisor: Jelte Bos, Location: TNO, Soesterberg

Virtual reality (VR) is a technology which is becoming more popular. A problem with this technology is the motion sickness that is associated with VR exposure, called cybersickness. A safety issue of cybersickness are the after-effects that can develop following the VR experience, such as disorientation and vertigo. Therefore, activities like driving an automobile are discouraged after VR exposure. The main cause the conflict between the visual and vestibular system. This could possibly be solved by adding an Earth-fixed reference frame (RF) that can reduce cybersickness. In this experiment, not only the effect of a RF will be investigated, but the effect of depth will also be researched. The RF will be placed in the fore- and background to see when the reducing effect on cybersickness is largest. We will measure the level of nausea people experience during the different conditions of the experiment. We hypothesize that by adding an Earth-fixed RF in the virtual world, it will reduce cybersickness and that this effect will be largest when it is in the background.

An investigation into the effect of multimedia learning principles, applied to the design of an animated instruction video for the online marketing tool ValueFlow.

Thesis

Rodee Voortman

Supervisor: Baptist Liefoghe, External supervisor: Astrid Lubeck (vacation) and Erik Prins, Location: Validators, Amsterdam

Moment-to-moment responses offer great opportunities for companies to analyze their advertisement material with the aid of real time affective reactions from consumers. Validators is a marketing and research company that has developed such a measurement tool, called ValueFlow. Participants have to continuously indicate their appreciation with a slider on the right, while watching a television commercial. ValueFlow is online software and includes an instructional animation video to explain what participants are required to do beforehand. However, Validators has observed unintended participant behaviour. Therefore, a new instruction has been designed with the objective to explore whether it can facilitate less unintended behaviour, compared to the current instruction. Additionally, the interaction between instruction and device is measured, to investigate whether results depend on type of device (mobile or computer). Instructional manipulations were made on a valid basis, considering Mayers' multimedia principles and theoretical aspects in the field of animation videos. Statistical ordinal regression has shown that the new instruction was effective for clicking/tapping behaviour. It is of interest for Validators to determine whether the new instruction guides participants to use ValueFlow as intended. More generally, the effects of the theoretical multimedia principles are tested in this research.

Anadiplosis and paragraph reading: An eye-tracking experiment

Thesis

Hector Mangas Afonso

Supervisor: Jessica Heeman, Location: Utrecht University

From a perceptual point of view, a paragraph is a very complex and tangled unit of information. They contain a multitude of shapes that can't be interpreted all at once. Navigation within these blocks of text is thus essential. Since early times, type workers have used different devices to improve such navigation. In the 15th century, a common practice was to add the first word of the next page at the bottom of the current one. The present study replicates this idea but within paragraphs, hypothetically to improve return sweep performance and overall readability. Two design choices for the repetition of words were considered (one embedded and one outside), as well as a graphic addition condition (diamonds before each line) and a control (no additions). Results showed no considerable improvement for the embedded additions, some limited improvement when the words were outside, and significant improvement when diamonds were present. This aligns with previous research, which suggests back sweep performance is linked with parafoveal perception.

Evaluating the my DMG MORI Customer Portal: How to increase Perceived Usability and the Clients' Usage.

Thesis

Alexa Horn

Supervisor: Dr. Martijn Mulder, External supervisor: Dr. Thomas Froitzheim, Location: DMG MORI, Bielefeld, Germany

DMG MORI is a global manufacturing company of, for example, lathes, metal cutting machines, milling machines, and industrial services. They want to further improve their digital service processes which includes the my DMG MORI customer portal. In addition, they want to get insights into the customer satisfaction of version 1 and 2 of the portal. Therefore, in this report semi-structured interviews with active and non-active users were done. Based on that, a questionnaire was developed to get additional data about consumer behaviour and satisfaction. Preliminary results of the interviews show that clients are in general satisfied with the performance of the customer portal. However, also points of improvement were mentioned. Data of the questionnaire still needs to be analysed, so results might change. First suggestions for future development are e.g., more detailed instructions of the functions and usage of my DMG MORI or the function of editing an already existing service request.

Does interactivity and teacher instruction help increase student motivation for engineering students who take service mathematics classes in TU Delft?

Winnie Dao

Supervisor: Leendert van Maanen, External supervisor: Jacqueline Wong, Location: TU Delft

Combined
Intern-
ship/Thesis

Research suggests that engineering undergraduates' lack of awareness about the importance of mathematics for their study success and for their careers contributes to their low motivation for mathematics. As student motivation is closely related to academic results, it is important that this problem is addressed. One way PRIME is attempting to address this via interactive videos on various topics to help students visualize the math. But does interactivity and teacher instruction help increase student motivation for engineering students who take service mathematics classes in TU Delft?

Evaluating the my DMG MORI Customer Portal: How to increase Perceived Usability and the Clients' Usage.

Maddy Koenraad

Supervisor: Irene van de Vijver, Location: Utrecht University

Thesis

Human cognition deteriorates as we age. Reinforcement learning (WL) - a fundamental cognitive function where we learn through trial and error - is one of those functions that declines. RL was thought to be manifested in the dopaminergic system. However, in recent years it has come to light that it, next to dopamine, also relies on working memory (WM). We use WM to temporarily hold critical information, and tend to compress extra information into WM through chunking, where we store items in groups instead of as single items. Here, we address the aging effect on the ability to chunk information. Differences in either conception, recognition and implementation of chunking strategies were studied. In addition, differences between visual and verbal stimuli were explored. Results show that ... (yet to be written)

Evaluating the my DMG MORI Customer Portal: How to increase Perceived Usability and the Clients' Usage.

Thesis

Bram de Bruijn

Supervisor: Stella Donker, External supervisor: Roalt Aalmoes and Naomi Sieben,
Location: Royal Netherlands Aerospace Centre

DMG MORI is a global manufacturing company of, for example, lathes, metal cutting machines, milling machines, and industrial services. They want to further improve their digital service processes which includes the my DMG MORI customer portal. In addition, they want to get insights into the customer satisfaction of version 1 and 2 of the portal. Therefore, in this report semi-structured interviews with active and non-active users were done. Based on that, a questionnaire was developed to get additional data about consumer behaviour and satisfaction. Preliminary results of the interviews show that clients are in general satisfied with the performance of the customer portal. However, also points of improvement were mentioned. Data of the questionnaire still needs to be analysed, so results might change. First suggestions for future development are e.g., more detailed instructions of the functions and usage of my DMG MORI or the function of editing an already existing service request.

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